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|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------|-------------|
| Programme      | B. Sc. Mathematics Honours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                           |                    |             |
| Course Code    | MAT7CJ401                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                           |                    |             |
| Course Title   | <b>MATHEMATICAL ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |                    |             |
| Type of Course | <b>Major</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |                    |             |
| Semester       | VII                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           |                    |             |
| Academic Level | 400-499                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                           |                    |             |
| Course Details | Credit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Lecture/Tutorial per week | Practicum per week | Total Hours |
|                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                         | 2                  | 75          |
| Pre-requisites | 1. Mathematical Logic and necessary exposure to set theory.<br>2. Basic Calculus<br>3. Real Analysis I, Real Analysis II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                           |                    |             |
| Course Summary | The topology of the real line is explored in detail, as is necessary later for an in-depth understanding of the theory of real functions. Limits, Continuity & Differentiation are rigorously covered. Riemann-Stieltjes Integration is introduced as a generalisation of the Riemann integration covered in earlier semesters, enabling the student to view summation of series and integration as extensions of the same concept. After a discourse on series of functions and various results discussing the compatibility of the above three notions with the limiting operations on series of functions, the course concludes with a presentation of the famous Stone-Weierstrass' Theorem. |                           |                    |             |

#### Course Outcomes (CO):

| CO                                                                                                                                                                                             | CO Statement                                                                                                                                                                                                                                                              | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|-------------------------------------------------------------|
| CO1                                                                                                                                                                                            | Analyse and differentiate between finite, countable, and uncountable sets, and apply these concepts to problems in $\mathbb{R}$                                                                                                                                           | An               | C                   | Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam |
| CO2                                                                                                                                                                                            | Evaluate the properties of compact, perfect, and connected sets in the context of metric spaces.                                                                                                                                                                          | E                | P                   | Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam |
| CO3                                                                                                                                                                                            | Synthesize the principles of continuity, differentiability, integrability and convergence of sequences and series including the application of the Mean Value Theorem and L'Hospital's Rule, to solve complex problems involving real-valued and vector-valued functions. | E                | P                   | Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                                                                                                                                                                                                                           |                  |                     |                                                             |

**Detailed Syllabus:**

| Textbook | Principles of Mathematical Analysis, Walter Rudin,, (3/e), McGraw Hill Inc(2013)                                                                                                                      |                                                                                                                                                                                       |                |                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------|
| Module   | Unit                                                                                                                                                                                                  | Content                                                                                                                                                                               | Hrs<br>(45+30) | External<br>Marks<br>(70) |
| I        | Basic Topology of the Real Line                                                                                                                                                                       |                                                                                                                                                                                       | 13             | Min.15                    |
|          | 1                                                                                                                                                                                                     | Chapter 2 – Finite, Countable & Uncountable Sets – 2.1 to 2.14                                                                                                                        |                |                           |
|          | 2                                                                                                                                                                                                     | Chapter 2 – Metric Spaces – 2.15 to 2.24                                                                                                                                              |                |                           |
|          | 3                                                                                                                                                                                                     | Chapter 2 – Metric Spaces – 2.25 to 2.30                                                                                                                                              |                |                           |
|          | 4                                                                                                                                                                                                     | Chapter 2 – Compact Sets – 2.31 to 2.42                                                                                                                                               |                |                           |
|          | 5                                                                                                                                                                                                     | Chapter 2 – Perfect Sets – 2.43 to 2.44                                                                                                                                               |                |                           |
|          | 6                                                                                                                                                                                                     | Chapter 2 – Connected Sets – 2.45 to 2.47                                                                                                                                             |                |                           |
| II       | Continuity and Differentiation                                                                                                                                                                        |                                                                                                                                                                                       | 16             | Min.20                    |
|          | 7                                                                                                                                                                                                     | Chapter 4 – Limits of Functions and Continuous Functions – 4.1 to 4.12                                                                                                                |                |                           |
|          | 8                                                                                                                                                                                                     | Chapter 4 – Continuity and Compactness – 4.13 to 4.21                                                                                                                                 |                |                           |
|          | 9                                                                                                                                                                                                     | Chapter 4 - Continuity and Connectedness – 4.22 to 4.24                                                                                                                               |                |                           |
|          | 10                                                                                                                                                                                                    | Chapter 4 – Discontinuities and Monotonic Functions – 4.25 to 4.30                                                                                                                    |                |                           |
|          | 11                                                                                                                                                                                                    | Chapter 5 – The Derivative – 5.1 to 5.6                                                                                                                                               |                |                           |
|          | 12                                                                                                                                                                                                    | Chapter 5 – Mean Value Theorems – 5.7 to 5.12                                                                                                                                         |                |                           |
|          | 13                                                                                                                                                                                                    | Chapter 5 – L’Hospital’s rule, Higher Derivatives & Taylor’s Theorem, Differentiation of Vector Valued Functions – 5.13 to 5.19 (proof of theorem 5.13 and theorem 5.15 are optional) |                |                           |
| III      | The Riemann-Stieltjes Integral                                                                                                                                                                        |                                                                                                                                                                                       | 9              | Min.15                    |
|          | 14                                                                                                                                                                                                    | Chapter 6 – Definition and Existence – 6.1 to 6.6                                                                                                                                     |                |                           |
|          | 15                                                                                                                                                                                                    | Chapter 6 – Definition and Existence – 6.6 to 6.11                                                                                                                                    |                |                           |
|          | 16                                                                                                                                                                                                    | Chapter 6 – Properties – 6.12 to 6.13                                                                                                                                                 |                |                           |
|          | 17                                                                                                                                                                                                    | Chapter 6 – Properties – 6.14 to 6.19 (proof of theorem 6.19 is optional)                                                                                                             |                |                           |
|          | 18                                                                                                                                                                                                    | Chapter 6 – Integration & Differentiation – 6.20 to 6.22                                                                                                                              |                |                           |
| IV       | Sequences & Series of functions                                                                                                                                                                       |                                                                                                                                                                                       | 7              | Min.10                    |
|          | 19                                                                                                                                                                                                    | Chapter 7 – Discussion of Main Problem - 7.1 to 7.3                                                                                                                                   |                |                           |
|          | 20                                                                                                                                                                                                    | Chapter 7 – Discussion of Main Problem - 7.4 to 7.6                                                                                                                                   |                |                           |
|          | 21                                                                                                                                                                                                    | Chapter 7 –Uniform Convergence – 7.7-7.10                                                                                                                                             |                |                           |
|          | 22                                                                                                                                                                                                    | Chapter 7 –Uniform Convergence & Continuity – 7.11 to 7.13                                                                                                                            |                |                           |
| V        | Practicum :<br>The goal is for the students to learn the following selected topics via self-study and group activities. The lecturer may assist by running and overseeing group discussions and class |                                                                                                                                                                                       | 30             | -                         |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--|--|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | seminars and referring library books for self-study and note preparation.                |  |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 3 – Convergent Sequences, Subsequences                                           |  |  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 3 – Cauchy Sequences, Upper and Lower Limits                                     |  |  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 3 – Some Special Sequences, Series                                               |  |  |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 3 – Series of Non-Negative Terms, The Root and Ratio Tests                       |  |  |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 3 – Power Series, Absolute Convergence                                           |  |  |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 3 – Addition and Multiplication of Series, Rearrangements.                       |  |  |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 4 – Infinite Limits & Limits at Infinity – 4.32 to 4.34                          |  |  |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 6 – Integration of Vector-valued Functions and Rectifiable curves - 6.23 to 6.27 |  |  |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Chapter 7 – Uniform Convergence, Integration and Differentiation – 7.16 to 7.18          |  |  |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Chapter 7 –Equicontinuity and Stone-Weierstrass Theorem – 7.19 to 7.27                   |  |  |
| <b>References</b> <ol style="list-style-type: none"> <li>1. Mathematical Analysis, T. M. Apostol, (2nd Edn.); Narosa; 2002.</li> <li>2. Introduction to Real Analysis, R. G. Bartle and D.R. Sherbert;; John Wiley Bros; 1982.</li> <li>3. Real Analysis- a first course, R. A. Gordon:(2nd Edn.); Pearson; 2009.</li> <li>4. Analysis-I, H. Amann and J. Escher, Birkhuser, 2006</li> <li>5. The way of Analysis, Robert Strichartz, (R/e), Jones and Bartlett Mathematics (2000)</li> <li>6. A first course in Real Analysis, M. H. Protter and C. B. Moray, Springer Verlag UTM (1977)</li> </ol> |                                                                                          |  |  |

**\*Optional topics are exempted for end semester examination**

**\*\*70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

**Mapping of COs with PSOs and POs :**

|      | PSO1 | PSO2 | PSO3 | PSO4 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|------|------|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| CO 1 | 3    | 2    | 1    | 0    | 3   | 0   | 3   | 0   | 3   | 0   | 0   |
| CO 2 | 2    | 3    | 2    | 0    | 3   | 0   | 3   | 0   | 3   | 0   | 0   |
| CO 3 | 3    | 3    | 3    | 1    | 3   | 0   | 3   | 0   | 3   | 0   | 0   |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics:**

|      | Internal Exam | Assignment/ Report | Seminar | Viva | End Semester Examinations |
|------|---------------|--------------------|---------|------|---------------------------|
| CO 1 | ✓             | ✓                  | ✓       | ✓    | ✓                         |
| CO 2 | ✓             | ✓                  | ✓       | ✓    | ✓                         |
| CO 3 | ✓             | ✓                  | ✓       | ✓    | ✓                         |